Arista Access Node BTM-ACC-19 QoS Congestion Event Resolution SOP

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5123

Description

This Standard Operating Procedure (SOP) is designed to address a minor severity QoS congestion event on the Arista Access node BTM-ACC-19, located at the BTM Layout Aggregation site. The congestion event is attributed to microbursts, interface tail drops, and QoS queue starvation. This failure domain occurs due to the inherent properties of the Arista Access node's architecture and the implementation of QoS policies. The Arista Access node employs a hierarchical QoS architecture, where traffic is classified, policed, and scheduled based on the DSCP values of the packets. However, in the presence of microbursts or sudden surges in traffic, the interface queues may become congested, resulting in tail drops and QoS queue starvation. The Arista Access node's hardware and software mechanisms, such as the 1G or 10G interface cards, the Arista EOS operating system, and the QoS policy configuration, all contribute to the occurrence of this failure domain.

Pre-requisite Checks

- Ensure that the BTM Layout Aggregation site is accessible and the remote hands are dispatched to the site.

- Verify that the redundant link is functioning correctly and is not the root cause of the congestion event.

- Check the power and cooling systems of the node to ensure that they are functioning within normal operating parameters.

Potential Root Causes

Cause	Probability
Microbursts or sudden surges in traffic	High
Incorrect QoS policy configuration	Medium
Hardware or software issues with the interface cards or Arista EOS	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: show interface status for BTM-ACC-19, Description: This command will display the operational status of the interface on BTM-ACC-19, including the current utilization, drop counts, and other relevant statistics.

Command: show qos policy, Expected Output: QoS policy configuration for BTM-ACC-19, Description: This command will display the QoS policy configuration for the access node, including the classification, policing, and scheduling rules.
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Command: show qos queue statistics, Expected Output: QoS queue statistics for BTM-ACC-19, Description: This command will display the QoS queue statistics for the access node, including the current queue depths, drop counts, and other relevant metrics.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy, Description: Verify that the QoS policy configuration for the access node is correct and does not contain any incorrect or missing rules.

Step 2: Step: Adjust QoS policy rules, Command: config qos policy classify policer , Description: Adjust the QoS policy rules to ensure that the policer is correctly configured to prevent microbursts or sudden surges in traffic.

Step 3: Step: Verify interface configuration, Command: show interface status, Description: Verify that the interface configuration for the access node is correct and does not contain any incorrect or missing settings.

Step 4: Step: Adjust interface settings, Command: config interface , Description: Adjust the interface settings to ensure that the interface is correctly configured to handle the expected traffic rates.

Post-Check Verification

```
Command: show qos queue statistics, Expected Output: QoS queue statistics for BTM-ACC-19, Description: Verify that the QoS queue statistics for the access node have returned to normal after the resolution steps were completed.

Command: show interface status, Expected Output: show interface status for BTM-ACC-19, Description: Verify that the interface status for the access node has returned to normal after the resolution steps were completed.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: rollback qos policy , Description: Restore the original QoS policy configuration for the access node in case the resolution steps failed to resolve the issue.

- Step: Restore original interface configuration, Command: rollback interface , Description: Restore the original interface configuration for the access node in case the resolution steps failed to resolve the issue.